

## Demo 7 — Mapping Color Patterns onto a Phylogeny

EEB 187: Ecology and Evolution of Color

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Week 8 Demo Section

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### Today's big question

**Do closely related fish species have similar color patterns?**

Or does natural selection override evolutionary history?

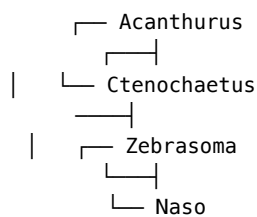
We'll answer this with data, a phylogeny, and two statistical tests.

### What is a phylogeny?

- A **family tree** for species
- Branch lengths = **time** (millions of years)
- Closely related species share a recent **common ancestor**
- The tree is estimated from DNA sequences

**Today's tree:** 65 surgeonfish species (*Acanthuridae*)

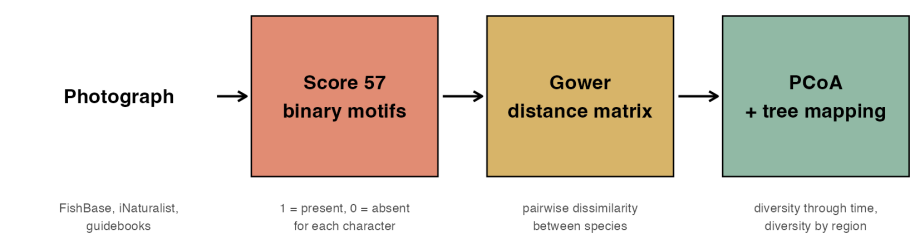
Root age: **65 million years** — right at the end-Cretaceous extinction!



(simplified — the real tree has 65 tips)

### Frédérich et al. 2026 — the paper at a glance

**Rapid and repeated evolution of pigmentation patterns in reef fishes.** Frédéric, Mittelheiser, Gillet, Hodge, Laudet & Dornburg. *BMC Biology* (2026) 24:46.  
<https://doi.org/10.1186/s12915-026-02544-4>



The scope

- **6 reef-fish families**
- **918 species** total
- **75%+** of extant diversity in every family

The coding

- **30 motifs** (stripes, spots, eyespots, marbling, beehive, saddle, ...)
- × **3 body regions** (head, trunk, tail)
- = a per-species binary fingerprint

The analyses

- Gower distance → PCoA
- Disparity-through-time (DTT)
- Blomberg’s K, OU vs BM vs AC
- Motif richness ↔ species richness

Today: the **Acanthuridae** sub-dataset → 83 species × 57 columns (the 30 motifs expand to 57 binary columns with “1 vs 2+” splits, eye-overlap variants, and 5 whole-body indices).

How they coded color — body regions × motifs

**Motif vocabulary** *(scored in each region)*

| Motif                                 | Description                                                 |
|---------------------------------------|-------------------------------------------------------------|
| <b>h-stripe / v-stripe / o-stripe</b> | horizontal / vertical / oblique lines (oblique = head only) |
| <b>stain</b>                          | large irregular patch                                       |
| <b>point</b>                          | small spot                                                  |
| <b>eyespot</b>                        | concentric ring (often a false eye)                         |
| <b>stripe/stain through eye</b>       | motif that overlaps the eye region                          |
| <b>marbling</b>                       | reticulate / vermiculate                                    |
| <b>beehive</b>                        | honeycomb / hexagonal                                       |
| <b>saddle</b>                         | curved band across tail                                     |
| <b>colored / stained scalpel</b>      | the Acanthuridae spine pigment                              |

HEAD 22 chars TRUNK 14 chars TAIL 17 chars

Every species = a row of 57 binary characters across these three zones

Each scored as **1 = present, 0 = absent**, with extra splits for **1 vs 2+** occurrences (e.g., H\_2\_h\_strips\_+ means 2 or more horizontal stripes on the head).

Three species, coded — the intuition



*Mostly one color + scalpel*



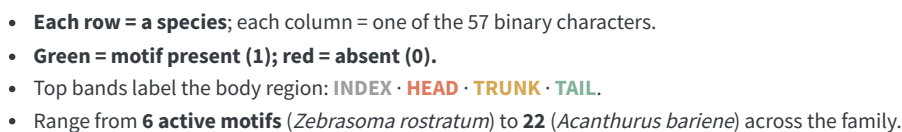
*Stripe signature on head, trunk, tail*



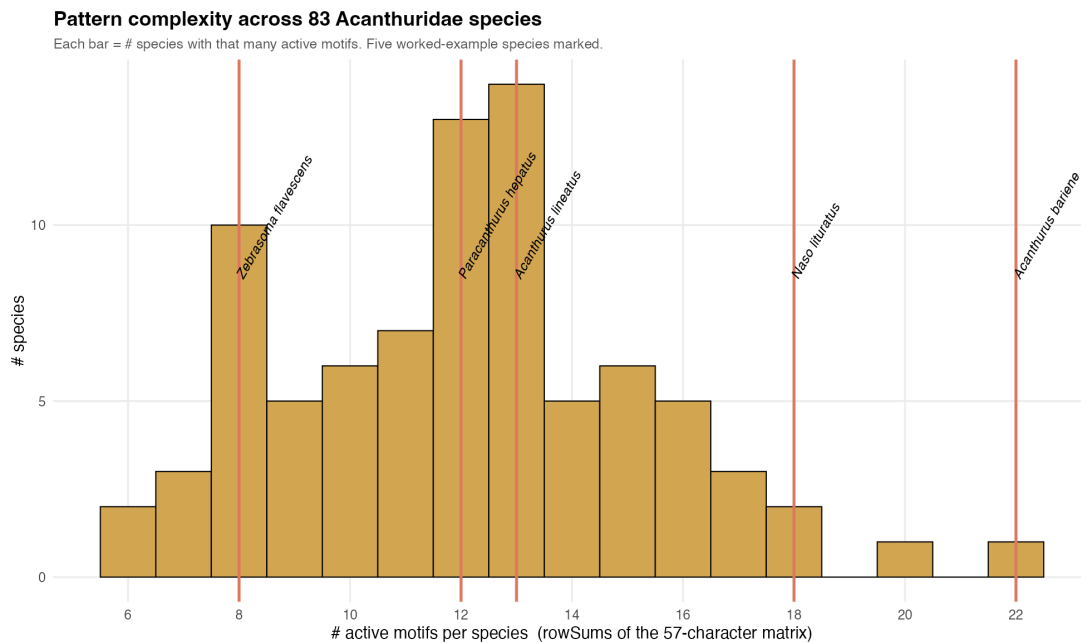
*Marbled face + colored scalpel + saddle*

### Five species, all 57 characters — the formal view

Five Acanthuridae species showing the range from minimal (Zebrasoma) to maximally encoded (A. bariene, 22 active motifs)



## Pattern complexity varies across the family



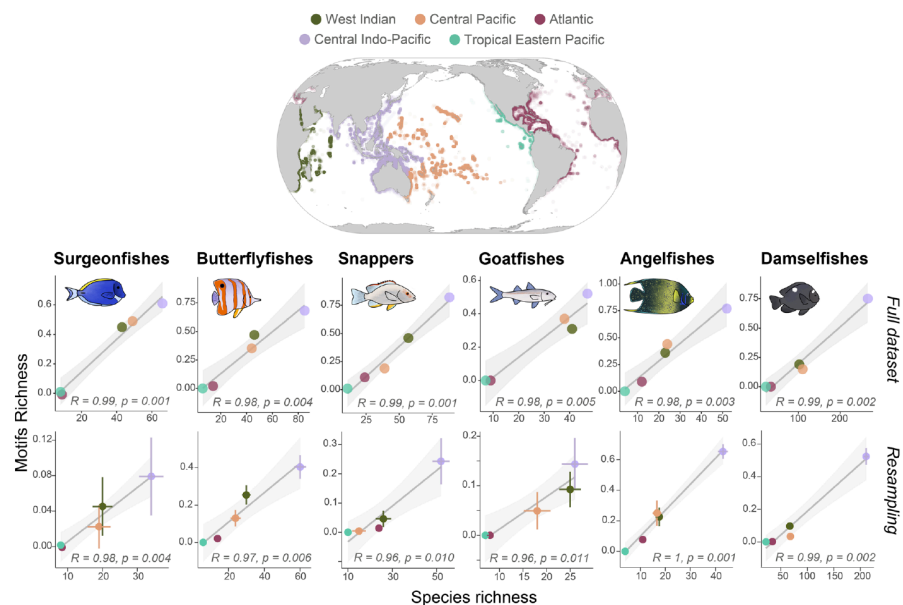
Distribution of motif counts across 83 Acanthuridae species. **Median = 12. Min = 6 (Zebrasoma rostratum). Max = 22 (Acanthurus bariene).**

This **single number per species** is what you'll run through **contMap**, Blomberg's K, and Pagel's  $\lambda$  in the worksheet.

## Result 1 — motif richness tracks species richness

Frédérich et al. BMC Biology (2026) 24:46

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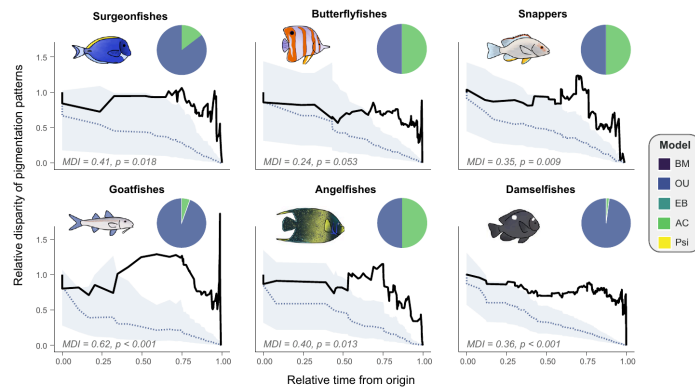


**Fig. 1** Relationships between motif richness and species richness from the five main biogeographic regions. The richness of pigmentation patterns (i.e., motif richness) is directly proportional to the number of species present in each region. Tests of correlation are all significant and Pearson's

- **Fig 1 from Frédéric et al. 2026.** Each panel = one family × 5 biogeographic regions. *Top row* = full dataset, *bottom row* = re-sampled to avoid widespread-species inflation.
- **Every family × every sampling regime:  $R > 0.96$ ,  $p < 0.01$ .** More species in a region → more motif diversity.
- **Implication:** speciation events drive pattern divergence — pattern diversity scales linearly with the species pool, not with the latitude/longitude of the reef.

## Result 2 — pattern evolution is rapid AND bounded

high level of homoplasy observed in animal coloration, diversification reflects highly rapid evolution within a



**Fig. 2** Disparity through time (DTT) plots illustrating the evolution of subclade disparity across the phylogeny of each fish family. Higher relative disparity values (y-axis) correspond to a greater average volume of pigmentation pattern space occupied by subclades relative to the disparity of the entire family. The solid line represents the observed disparity calculated for each clade, while dashed line is median expected disparity under a null Brownian motion model based on simulations. Across all fish families, subclade disparity was significantly higher than expected under BM, with sharp increases of within subclade disparity toward the present, suggesting the potential for high motif convergence between subclades. Inset pie charts display the distribution of the Akaike information criterion weights (AICw) from trait evolutionary model (BM, OU, EB, AC, and psi) comparisons. Generally, support for BM, EB, and psi models was too low to be discernible on pie charts. Plots reflect model comparisons for PC1. Plots of PCs 2–3 are provided in Additional file 2: Fig. S2 and Additional file 2: Fig. S3.

### Phylogenetic signal is weak in every family

(multivariate Blomberg's K, all  $p < 0.01$ ):

| Family         | K_mult |
|----------------|--------|
| Acanthuridae   | 0.11   |
| Chaetodontidae | 0.10   |
| Lutjanidae     | 0.22   |
| Mullidae       | 0.08   |
| Pomacanthidae  | 0.05   |
| Pomacentridae  | 0.06   |

- Best-fit evolutionary model: **OU1** (single optimum, bounded morphospace) + **AC** (accelerating rate).
- **Phylogenetic half-life**  $\approx 1.5$  Myr vs  $\sim 37.5$  Myr for other fish morphological traits.

**Punchline:** patterns evolve fast within a constrained set of motifs  $\rightarrow$  strong convergence + recurring labyrinths, stripes, spots across distantly related lineages.

## What is phylogenetic signal?

The tendency for **closely related species to resemble each other** more than distantly related species.

### Strong signal

Relatives look alike  $\rightarrow$  color patterns evolve slowly

*Pattern is "conserved" on the tree*

### Weak/no signal

Relatives look different  $\rightarrow$  color changes fast, convergence is common

*Pattern evolves independently of relatedness*

We measure this with two statistics: **Blomberg's K** and **Pagel's  $\lambda$**

## Blomberg's K — what the number means

| K value                | Interpretation                                                |
|------------------------|---------------------------------------------------------------|
| <b>K = 0</b>           | No signal — trait is random on the tree                       |
| <b>0 &lt; K &lt; 1</b> | Signal exists, but weaker than expected under Brownian motion |
| <b>K = 1</b>           | Trait matches Brownian motion prediction exactly              |
| <b>K &gt; 1</b>        | Relatives are MORE similar than Brownian motion predicts      |

The **p-value** tells you if the signal is statistically significant:

- **$p < 0.05$**   $\rightarrow$  yes, there's real signal
- **$p > 0.05$**   $\rightarrow$  can't distinguish from random

**K = 0.16 with p = 0.04** → “there IS signal, but it’s modest — color patterns show some phylogenetic conservatism but also a lot of independent change”

## contMap — painting traits on the tree

**contMap** reconstructs ancestral trait values and paints them along every branch.

- Cool colors (blue) = **simple** patterns
- Warm colors (red) = **complex** patterns
- You can SEE where on the tree complexity changed
- Look for **abrupt shifts** — those are evolutionary events

You’ll generate one of these in Section 5 of the worksheet.

The most complex species? *Acanthurus bariene* — 19 pattern elements!

The simplest? *Zebrasoma rostratum* — only 4 elements.

## Discrete traits — mapping presence/absence

Not all traits are continuous. Some are **binary**: the species either has it or doesn’t.

**Example: trunk marbling** (irregular worm-like markings)

- 17 of 62 species have trunk marbling
- Concentrated in *Acanthurus* (11/26) and *Ctenochaetus* (3/6)
- Absent in *Zebrasoma*, *Prionurus*, *Paracanthurus*

**Discussion question:**

Is marbling a Turing pattern?

Think back to Lecture 12 — reaction-diffusion produces labyrinthine patterns that look a lot like “marbling.”

You’ll color tree tips by marbling presence and ask: how many times did it evolve?

## The workflow — what you’ll do today

| Step | Section | What you do              | What you learn                           |
|------|---------|--------------------------|------------------------------------------|
| 1    | \$1–2   | Load tree + plot it      | How a phylogeny looks in R               |
| 2    | \$3     | Load color data          | What coded pattern traits look like      |
| 3    | \$4     | Match tree ↔ data        | The “golden rule” of comparative methods |
| 4    | \$5     | Create complexity score  | Summarizing many traits into one number  |
| 5    | \$6     | contMap                  | Painting a trait on the tree             |
| 6    | \$7     | Blomberg’s K + Pagel’s λ | Testing phylogenetic signal              |
| 7    | \$8     | Map trunk marbling       | Discrete traits on a tree                |
| 8    | \$9     | Pick your own trait      | Independent analysis                     |

~60 minutes of guided work, then 15 minutes on your own

## What you’ll turn in

1. **Screenshot** of your contMap (pattern complexity on the tree)

2. **Your  $K$  and  $\lambda$  values** with a one-sentence interpretation each
3. **Your chosen trait** from “Your Turn” — which trait did you pick? What was  $K$ ? Was it significant?
4. **One paragraph**: “Is color pattern diversity phylogenetically conserved in surgeonfishes? Why or why not?”

There is no single “right answer” — we want a **nuanced** interpretation backed by your data.

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## Getting started

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1. Open **RStudio**
2. Set your working directory:

```
setwd("path/to/demo-labrid-pcm-week8")
```

3. Open the worksheet: `demo7-worksheet.qmd` (or the HTML version in your browser)
4. Work through it **section by section** — don’t skip ahead
5. **Talk to your neighbor**. If you’re stuck, ask. If you’re not stuck, explain what you’re seeing.

**Let’s go!**